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EFFICIENT ESTIMATION USING PANEL DATA

BY TREVOR S. BREUSCH, GRAYHAM E. MIZON, AND PETER SCHMIDT¹

1. INTRODUCTION

IN AN IMPORTANT RECENT PAPER, Hausman and Taylor (1981)—hereafter HT—considered the instrumental-variable estimation of a regression model using panel data, when the individual effects may be correlated with a subset of the explanatory variables. They provided a simple consistent estimator and an efficient estimator. More recently, Amemiya and MaCurdy (1986)—hereafter AM—have suggested an alternative estimator which is more efficient than the HT estimator, under certain conditions and given stronger assumptions than HT made. However, the relationship between the HT and AM papers is less clear than it might be, in part because of notational differences between the two papers. In this paper we clarify the relationship between the HT and AM estimators, and we show that the difference between these estimators lies in the treatment of the time-varying explanatory variables which are uncorrelated with the effects: HT use each such variable as two instruments (means and deviations from means), while AM use such variables as $T + 1$ instruments (as deviations from means and also *separately* for each of the T available time periods). This enables us to make clear the conditions under which the AM estimator is more efficient than the HT estimator. We also present each estimator in a form which allows it to be calculated using standard instrumental-variables (two-stage least squares) software.

Following the AM path one step further, we then define a third (BMS) estimator which, under yet stronger assumptions, is more efficient than the AM estimator. Both HT and AM use as instruments the deviations from means of the time-varying variables which are correlated with the effects. A more efficient estimator may be obtained by using *separately* the $(T - 1)$ linearly independent values of these deviations from individual means. Consistency requires that these be legitimate instruments, and whether this is so depends on why these time-varying variables are correlated with the effects. For example, if such correlation arises solely because of a time-invariant component which is removed in taking deviations from individual means, these instruments are legitimate.

2. THE HT ESTIMATOR

We consider the same model as HT, and use more or less their notation. The model is

$$(1) \quad y_{it} = X'_{it}\beta + Z'_i\gamma + \alpha_i + \varepsilon_{it} \quad (i = 1, \dots, N; t = 1, \dots, T),$$

or, in matrix form,

$$(2) \quad y = X\beta + Z\gamma + \alpha + \varepsilon.$$

The errors ε_{it} are iid $N(0, \sigma_\varepsilon^2)$ while the individual effects α_i are iid $N(0, \sigma_\alpha^2)$. The errors ε_{it} are uncorrelated with X_{it} , Z_i , and α_i , while the effects α_i may be correlated with some of the explanatory variables. We note explicitly that Z_i is time-invariant, while X_{it} varies over both i and t .

We use the following notation for projections. For any matrix A , let P_A be the projection onto the column space of A ; thus $P_A = A(A'A)^{-1}A'$ if A is of full column rank.

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Let $Q_A = I - P_A$ be the projection onto the space orthogonal to A . Let V be the $NT \times N$ matrix of individual dummy variables, so that P_v converts an NT vector like y into a vector of individual means, while Q_v converts it into deviations from individual means. Note that, for the time-invariant variables Z , $P_v Z = Z$ and $Q_v Z = 0$, while for the time-varying variables X , $X = Q_v X + P_v X$, where $(Q_v X)'(P_v X) = 0$ and $[(Q_v X), (P_v X)]$ is assumed to have full column rank. Also as a matter of notation, let $\theta^2 = \sigma_\epsilon^2 / (\sigma_\epsilon^2 + T\sigma_\alpha^2)$; and note that

$$(3) \quad \text{cov}(\alpha + \epsilon) = \sigma_\epsilon^2 \Omega, \quad \text{where} \quad \Omega^{-1} = Q_v + \theta^2 P_v, \quad \Omega^{-1/2} = Q_v + \theta P_v.$$

Finally, again following HT, we partition X and Z :

$$(4) \quad X = (X_1, X_2), \quad Z = (Z_1, Z_2),$$

where X_1 and Z_1 are uncorrelated with α (e.g., $X_1' \alpha / NT \rightarrow 0$, where “ $\rightarrow$ ” indicates convergence in probability) but X_2 and Z_2 are correlated with α . Their dimensions are denoted as follows: $k = k_1 + k_2$, $g = g_1 + g_2$, where X_1 is $NT \times k_1$, etc. We will refer to X_1 and Z_1 as “exogenous” and to X_2 and Z_2 as “endogenous.” It is assumed that X does not contain any lagged values of y , so that the additional complications found in dynamic panel data models (see, e.g., Bhargava and Sargan (1983)) are avoided.

We are now in a position to define the HT efficient estimator. First transform (1) by $\Omega^{-1/2}$ to make the error term have a scalar covariance matrix:

$$(5) \quad \Omega^{-1/2} y = \Omega^{-1/2} X \beta + \Omega^{-1/2} Z \gamma + \Omega^{-1/2} (\alpha + \epsilon).$$

Next, perform IV where the list of instruments is $A = (Q_v, X_1, Z_1)$. This defines the HT efficient estimator, say $\tilde{\beta}_{HT}, \tilde{\gamma}_{HT}$.

Since A is not of full column rank, the usual IV formulae involving $(A'A)^{-1}$ do not apply. However, if we let $H = (X_1, Z_1)$, and $A = (Q_v, H)$, then

$$(6) \quad P_A = Q_v + P_{(P_v H)} = Q_v + P_v H (H' P_v H)^{-1} H' P_v.$$

Thus the projection of a variable onto A equals its deviations from (individual) means *plus* its projection (or the projection of its means) onto the means of (X_1, Z_1) .

We now show that the HT estimator can be rewritten in such a way that it can be calculated using standard IV (2SLS) software.

THEOREM 1: *The HT efficient estimator $(\tilde{\beta}_{HT}, \tilde{\gamma}_{HT})$ is IV of (5) using as instruments either $A = (Q_v, X_1, Z_1)$, or*

$$B = (Q_v X, X_1, Z_1) = (Q_v X_1, Q_v X_2, X_1, Z_1) \quad \text{or}$$

$$C = (Q_v X, P_v X_1, Z_1) = (Q_v X_1, Q_v X_2, P_v X_1, Z_1).$$

PROOF: It is clear that $P_B = P_C$, since the column space of $(Q_v X_1, X_1)$ is the same as the column space of $(Q_v X_1, P_v X_1)$ —and, indeed, is the same as the column space of $(X_1, P_v X_1)$. Thus B and C are equivalent instrument sets and lead to the same estimator.

On the other hand, it is not true that $P_B = P_A$. Indeed, P_B is (strictly) contained in P_A . However, for the (transformed) regressors in (5), the projections onto A and onto B are

the same, so the same estimator results. Specifically, these projections are as follows:

$$\begin{aligned}
 P_B \Omega^{-1/2} X_1 &= P_B (Q_v X_1 + \theta P_v X_1) \\
 &= Q_v X_1 + \theta P_v X_1 = \Omega^{-1/2} X_1 = P_A \Omega^{-1/2} X_1, \\
 P_B \Omega^{-1/2} Z_1 &= \theta Z_1 = P_A \Omega^{-1/2} Z_1, \\
 P_B \Omega^{-1/2} X_2 &= P_B (Q_v X_2 + \theta P_v X_2) \\
 &= Q_v X_2 + \theta P_{(P_v X_1, Z_1)} (P_v X_2) = P_A \Omega^{-1/2} X_2, \\
 P_B \Omega^{-1/2} Z_2 &= P_B \theta Z_2 = \theta P_{(P_v X_1, Z_1)} Z_2 = P_A \Omega^{-1/2} Z_2.
 \end{aligned}$$

This result is of interest for three reasons. First, instrument set B (or C) is of full column rank, so that the HT estimator can now be calculated using standard IV (two stage least squares) software. Second, the HT order condition for the existence of the estimator emerges naturally as the requirement that there be as many instruments as regressors:

$$2k_1 + k_2 + g_1 \geq k_1 + k_2 + g_1 + g_2, \quad \text{or} \quad k_1 \geq g_2.$$

Third, the HT efficiency argument (p. 1387) can be questioned because of an incidental parameters problem in their reduced form equations for X_2 and Z_2 : the dimensions of their π_{13} and π_{23} (coefficients of Q_v in the X_2 and Z_2 equations) increase with sample size. However, there is no longer an incidental parameters problem in the HT reduced form if we replace Q_v by $(Q_v X)$ in their (3.4).

3. THE AM ESTIMATOR

We now define an estimator which, if it is consistent, is no less efficient than the HT efficient estimator. To do so, we need to define a notational convention. Suppose that S_{it} is any $1 \times L$ (row) vector of time-varying variables, and that S is the corresponding $NT \times L$ data matrix (displayed below). Then S^* is defined to be the $NT \times LT$ matrix:

$$(7) \quad S = \begin{bmatrix} S_{11} \\ \vdots \\ S_{1T} \\ \hline \vdots \\ \vdots \\ S_{NT} \end{bmatrix}, \quad S^* = \begin{bmatrix} S_{11} & S_{12} & \cdots & S_{1T} \\ \vdots & \vdots & & \vdots \\ S_{11} & S_{12} & \cdots & S_{1T} \\ \hline \vdots & \vdots & & \vdots \\ S_{N1} & S_{N2} & \cdots & S_{NT} \\ \hline \vdots & \vdots & & \vdots \\ S_{N1} & S_{N2} & \cdots & S_{NT} \end{bmatrix}.$$

The essential point is that each column of S contains values of S_{it} for all values $t = 1, 2, \dots, T$, while each column of S^* contains values of S_{it} for *one* t only. Note S^* is “time-invariant” in the HT sense: $Q_v S^* = 0$, $P_v S^* = S^*$.

Now define $(\tilde{\beta}_{AM}, \tilde{\gamma}_{AM})$ as the IV estimator of (5), using instruments $A^0 = (Q_v, X_1^*, Z_1)$. This is essentially the AM estimator $\hat{\alpha}_0$ (their equation (3.9)) translated into our notation. (We say “essentially” because AM model 3 does not contain time-invariant exogenous variables.) The translation involves straightforward algebra and can be found in Breusch, Mizon, and Schmidt (1987, Appendix 2).

This discussion presumes that every variable in X_1 varies over i as well as t . If any variable in X_1 varies only over t , it can be used only as one instrument, rather than as T instruments as in AM.

The consistency of the AM estimator requires a stronger exogeneity assumption for X_1 than does the consistency of the HT estimator. HT require only the means of the variables in X_1 to be uncorrelated with the effects, whereas AM require uncorrelatedness separately at every point in time. As AM argue, however, it is hard to think of cases where HT's exogeneity condition would hold but AM's would not. In addition, the AM condition is required if the HT estimator is to remain consistent when only subsets of the time periods $1, \dots, T$ are used in estimation.

In Section 2 it was convenient to rewrite the HT estimator as an IV estimator using $Q_v X$ instead of Q_v (Theorem 1). The same is true here.

THEOREM 2: *The AM estimator $(\tilde{\beta}_{AM}, \tilde{\gamma}_{AM})$ is IV of (5), using as instruments $A^0 = (Q_v, X_1^*, Z_1)$, or*

$$B^0 = (Q_v X, X_1^*, Z_1) = (Q_v X_1, Q_v X_2, X_1^*, Z_1), \quad \text{or}$$

$$C^0 = [Q_v X, P_v X_1, (Q_v X_1)^*, Z_1] = [Q_v X_1, Q_v X_2, P_v X_1, (Q_v X_1)^*, Z_1].$$

PROOF: The proof that A^0 and B^0 lead to the same estimator is essentially the same as the proof of Theorem 1, and is therefore omitted. Furthermore, for any panel data matrix S , the projection onto S^* is the same as the projection onto $[(P_v S), (Q_v S)^*]$, since the means of S and any $T-1$ deviations from means determine (linearly) the T separate values of S , and conversely. The equivalence of B^0 and C^0 simply uses this fact for $S = X_1$.

The formulation of the AM estimator as IV with instrument set B^0 is useful for several reasons. First, B^0 is of full column rank, so that standard IV programs can be used. Second, by counting instruments and parameters to be estimated we easily arrive at the order condition for existence of the estimator, namely $Tk_1 \geq g_2$. Third, we can see clearly the difference between the HT and AM estimators, which lies in the treatment of the time-varying exogenous variables. HT use each such variable as two instruments ($Q_v X_1$ and $P_v X_1$) whereas AM use each such variable as $T+1$ instruments ($Q_v X_1$ and X_1^*).

The AM estimator will be more efficient than the HT estimator to the extent that (in the population) more of $\Omega^{-1/2} X_2$ and $\Omega^{-1/2} Z_2$ is explained by $B^0 = (Q_v X, X_1^*, Z_1)$ than is explained by $B = (Q_v X, X_1, Z_1)$. If we write formal reduced form equations for X_2 and Z_2 , with $Q_v X$, X_1^* and Z_1 as explanatory variables, the two estimators are equally efficient asymptotically if the coefficients of the variables in X_1^* are the same for all t . This differs from the condition given by AM, whose reduced form equations omit $Q_v X$ (or Q_v). In the reduced form equation (3.4) assumed by HT (p. 1386), this condition is assumed; the HT efficient estimator is efficient, given their assumed reduced form.

The difference between the HT and AM estimators can also be seen by comparing the HT instrument set C in Theorem 1 to the AM instrument set C^0 in Theorem 2. The HT and AM estimators both use $Q_v X_1$, $Q_v X_2$, $P_v X_1$ and Z_1 as instruments, but the AM estimator uses the *additional* instruments $(Q_v X_1)^*$. Note that $(Q_v X_1)^*$ has Tk_1 columns, but its rank is only $(T-1)k_1$, since for each variable only $(T-1)$ deviations from means are (linearly) independent. Therefore the matrix C^0 does not have full column rank, but it (generally) will if we use only $(T-1)$ deviations from means in $(Q_v X_1)^*$, for each time-varying exogenous variable.

4. A MORE EFFICIENT AM-LIKE ESTIMATOR

In this section we define another estimator, which uses even more instruments than the AM estimator. If these instruments are legitimate, this estimator will be efficient relative to the AM estimator.

To motivate this estimator, we note that the AM estimator differs from the HT estimator only in its treatment of the time-varying exogenous variables (X_1). In particular, the AM estimator treats the time-varying endogenous variables (X_2) exactly as the HT estimator does, using $Q_v X_2$ but not $P_v X_2$ as instruments. While it is obvious that $P_v X_2$ cannot be used as instruments, we can consider using $(Q_v X_2)^*$ as instruments, thus extending to X_2 the AM treatment of X_1 (except that $P_v X_2$ is not used). We therefore define the new estimator ($\tilde{\beta}_{BMS}, \tilde{\gamma}_{BMS}$) as IV of (5), using the instrument set

$$(8) \quad D^0 = [Q_v X_1, P_v X_1, (Q_v X_1)^*, Q_v X_2, (Q_v X_2)^*, Z_1] \\ = [Q_v X, P_v X_1, (Q_v X)^*, Z_1].$$

As just noted, this estimator differs from the AM estimator by its use of the $(T-1)k_2$ additional instruments in $(Q_v X_2)^*$. If these additional instruments are legitimate, the BMS estimator is at least as efficient as the AM estimator. It will be more efficient than the AM estimator if (in the population) more of $\Omega^{-1/2} X_2$ and $\Omega^{-1/2} Z_2$ is explained by D^0 than by C^0 . If we write formal reduced form equations for X_2 and Z_2 , with the variables in D^0 as explanatory variables, the AM and BMS estimators will be equally asymptotically efficient if the coefficients of $(Q_v X_2)^*$ equal zero. (If, in addition, the coefficients of $(Q_v X_1)^*$ equal zero, the HT estimator is also equally asymptotically efficient.)

The conditions under which the HT or AM estimators are efficient are testable, at least in principle, since they are just sets of linear restrictions on the reduced form equations. It may therefore be reasonable to test rather than assume these restrictions. On the other hand, if the variables in X_1 and X_2 are highly correlated over time, as they may be in many applications, the coefficients of the variables in $(Q_v X)^*$ may be estimated rather imprecisely, and the tests of these conditions may have very little power. Similar considerations apply to the question of how large the efficiency gain of the AM or BMS estimators over the HT estimator is likely to be. For a given sample, it is observable whether the use of their extra instruments increases the explanatory power of the reduced form equations substantially. This will naturally depend on the data set and the context, and is therefore a suitable subject for the empirical investigation. A recent paper (Cornwell and Rupert (1988)) reports a wage equation for which the standard error of the coefficient of education is .078 for HT, .059 for AM, and .031 for BMS, so that in at least one instance the extra instruments of AM and BMS make a noticeable difference in the (estimated) efficiency of estimation.

The question of whether $(Q_v X_2)^*$ is a legitimate set of instruments depends on what we assume about the nature of the correlation between X_2 and the effects. The effects are time-invariant, and the HT definition of correlation between X_2 and the effects is simply that the individual *means* of the variables in X_2 are correlated with the effects; thus $P_v X_2$ cannot be used as instruments. On the other hand, $(Q_v X_2)^*$ may or may not be correlated with the effects, depending on what we assume in the reduced form equation for X_2 . In particular, it is possible that X_2 is correlated with the effects only because it contains a time-invariant component which is correlated with the effects. If so, then $Q_v X_2$ does not contain this component and $(Q_v X_2)^*$ is legitimate. On the other hand, if the part of X_2 correlated with the effects is not time-invariant, then $(Q_v X_2)^*$ is not legitimate. ($Q_v X_2$ is still legitimate, of course, since Q_v is orthogonal to the effects.)

As an example of the issue involved, consider a typical potential application for these techniques, such as a wage equation in which X_2 is schooling. We worry about bias in the OLS estimates because ability affects schooling. If the effect of ability on schooling is

time-invariant, then deviations of schooling from the individual mean values are (separately) legitimate instruments. This may seem an obviously unreasonable assumption, since there is no apparent reason to believe that the effect of ability on schooling is time-invariant. On the other hand, one might argue that this is no more unreasonable than the assumption, already made in the model, that the effect of ability on wage (the individual effect) is time-invariant.

The hypothesis that the instruments in $(Q_v X_2)^*$ are legitimate is testable, of course. A simple (though not necessarily the most appropriate—see Holly (1982)) possibility is a Hausman-test of the difference between the AM and BMS estimators. In the Cornwell and Rupert (1988) wage equation, for example, Hausman tests fail to reject the legitimacy of the additional AM and BMS instruments.

5. CONCLUDING REMARKS

In this paper we have compared estimators proposed by HT and AM, and have proposed a third (BMS) estimator. As we progress from the HT to the AM and then to the BMS estimator, we require successively stronger exogeneity assumptions, and we achieve successively more efficient estimators. These exogeneity assumptions are testable. Nevertheless, it is somewhat disconcerting to have the choice of estimators depend on the properties of reduced form equations that are more or less devoid of behavioral content. Consider, for example, the (typical) application consisting of a wage equation, in which the unobservable effects are called ability, and one of the explanatory variables is schooling, which may be correlated with ability. All of the estimators considered in this paper arise from completing the system with a fairly arbitrary reduced form equation for schooling. A more standard procedure (from the point of view of the simultaneous equations literature, anyway) would be to complete the system with a structural (behavioral) schooling equation. In this case the choice of instruments would follow automatically; and, if the schooling equation is overidentified, joint estimation of the wage and schooling equations would be more efficient than estimation of the wage equation alone.

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